

Review Article

Histidine-rich glycoprotein: A novel adaptor protein in plasma that modulates the immune, vascular and coagulation systems

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Summary Histidine-rich glycoprotein (HRG) is an abundant plasma glycoprotein that has a multidomain structure, interacts with many ligands, and has been shown to regulate a number of important biological processes. HRG ligands include Zn^{2+} and haem, tropomyosin, heparin and heparan sulphate, plasminogen, plasmin, fibrinogen, thrombospondin, IgG, Fc γ R and complement. In many cases, the histidine-rich region of the molecule enhances ligand binding following interaction with Zn^{2+} or exposure to low pH, conditions associated with sites of tissue injury or tumour growth. The multidomain nature of HRG indicates that it can act as an extracellular adaptor protein, bringing together disparate ligands, particularly on cell surfaces. HRG binds to most cells primarily via heparan sulphate proteoglycans, binding which is also potentiated by elevated free Zn^{2+} levels and low pH. Recent reports have shown that HRG can modulate angiogenesis and additional studies have shown that it may regulate other physiological processes such as cell adhesion and migration, fibrinolysis and coagulation, complement activation, immune complex clearance and phagocytosis of apoptotic cells. This review outlines the molecular, structural, biological and clinical properties of HRG as well as describing the role of HRG in various physiological processes.

Key words: angiogenesis, histidine–proline-rich glycoprotein, histidine-rich glycoprotein, inflammation, tumour metastasis.

Introduction

Histidine rich glycoprotein (HRG), a single polypeptide chain α_2 -plasma glycoprotein of approximately 75 kDa, was first isolated in 1972.^{1,2} HRG is found in the plasma of many vertebrates, including humans, mice, rabbits, rats, chickens and cows. It is synthesized in liver parenchymal cells,^{3–5} although some studies suggest that HRG may also be produced by immune cells such as monocytes and macrophages,⁶ and by megakaryocytes.⁷ The molecule is present in human plasma at the relatively high concentration of 100–150 μ g/mL (~ 1.5 μ mol/L);^{5,8,9} plasma HRG levels at birth are approximately 20% that of adults, and levels gradually increase with age.⁵ Clinical studies have shown that families with congenital HRG deficiencies (20–35% of normal plasma levels) fail to show any abnormalities in routine laboratory assays of haemostatic and immunological function, suggesting that a plasma HRG level 20% of normal is probably sufficient for physiological function.¹⁰ Interestingly, HRG is one of the few plasma proteins reported to show a marked reduction (approximately 50%) during the last trimester of pregnancy.¹¹

HRG acts as a negative acute phase reactant whereby plasma levels are markedly reduced during the acute non-specific response to tissue injury.^{12,13} Thus, levels of plasma HRG have been found to decrease during acute states of

disease such as advanced liver cirrhosis,^{9,14} AIDS, renal disease, asthma and pulmonary disease.¹⁵ High plasma HRG levels have also been associated with thrombotic disorders.^{16,17} HRG can interact with different cell surface receptors, bind a wide range of ligands, and it has been implicated in the regulation of numerous biological functions, notably angiogenesis, immune complex clearance and coagulation. Many biological properties of HRG have been thoroughly investigated, and this review summarizes the molecular, structural, biological and clinical properties of HRG as well as reviewing the role of HRG in various physiological processes.

HRG protein structure

HRG was first isolated and characterized from human serum by Heimburger *et al.*² and was subsequently isolated and characterized from human plasma^{18,19} and platelets.^{7,20} Both plasma and platelet HRG appear immunohistochemically identical.⁷ The primary structure of human HRG is predicted to be a 507 amino acid³ multidomain polypeptide consisting of two cystatin-like regions at the N-terminus, a histidine-rich region (HRR) flanked by proline-rich regions (PRR), and a C-terminal domain (Fig. 1). The modular domain structure of HRG, the location of internal disulphide bonds and predicted carbohydrate attachment sites are represented as a schematic diagram in Figure 1. Perhaps the most striking feature of HRG is its unusually high content of histidine and proline residues, each constituting approximately 13% of total amino acids, which are concentrated within the HRR and PRR.^{1,18,21} The high content of both histidine and proline residues has resulted in HRG also being termed ‘histidine–proline-rich

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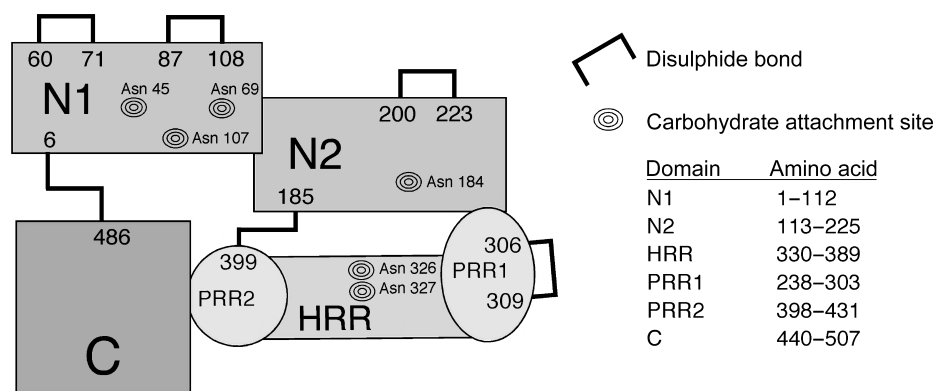


Figure 1 Modular structure of histidine-rich glycoprotein (HRG). Human HRG is comprised of three principal domains; namely, an amino-terminal domain (N1 and N2 regions), a central histidine-rich region (HRR) and a C-terminal domain. The HRR is flanked by two proline-rich sequences (PRR). Two internal disulphide bonds link the domains (represented by black solid lines); residue 6 in the N1 region links to residue 486 in the C-terminal domain and residue 185 in the N2 region links to residue 399 in PRR2. There are four intradomain disulphide bonds: between residues 60 and 71 in the N1 region, residues 87 and 108 in the N1 region, residues 200 and 223 in the N2 region and possibly residues 306 and 308 in PRR1. These disulphide bonds render various domains of the molecule resistant to proteolytic cleavage. In most humans, six glycosylation sites are conserved (HRG form-1), with three being present in the N1 region, one in the N2 region and two in the HRR. A polymorphism exists whereby Ser186 is substituted with Pro186, resulting in the loss of a glycosylation site (Asn184) in the N2 region (HRG form-2). The amino acid residues that comprise the different domains of the human HRG molecule are listed in the figure.

glycoprotein' (HPRG).²² Human HRG is also composed of approximately 14% carbohydrate attached to six predicted N-linked glycosylation sites (Fig. 1).^{1,3}

The amino acid sequence of human HRG contains internal repeat sequences that can be classified into five different types (I–V). Type V repeats include 12 tandem repeats of five consensus amino acids.³ The striking repetition of the consensus sequence is highlighted when the 12 tandem repeats of the pentapeptide sequence (between amino acids 330 and 389) are arranged in blocks (Fig. 2A). Of the 12 tandem amino acid repeats, the last five repeated sequences are identical, representing exact repetitions of a consensus GHHPH motif. It is likely this region evolved via successive duplication of a short stretch of 15 nucleotides (1328–1342) encoding for the GHHPH motif.³

Histidine residues are highly concentrated in the small region between amino acids 330 and 389 forming the 'histidine-rich' region of the protein.³ Proline residues are more widely distributed, although many are concentrated in the 'proline-rich' region immediately before and after the HRR. These PRR and HRR contain very few hydrophobic or basic amino acids and are predicted to form a structural framework supporting a twisted and elongated histidine-rich core.³ Amino acid sequence comparison between the HRR of HRG and HRR of high molecular weight kininogen reveals approximately 50% identity. The isoelectric point of HRG is approximately 6.45, suggesting that at physiological pH, HRG remains negatively charged. Amino acid sequence comparison between PRR of HRG and proline-rich proteins and peptides^{23–25} from parotid saliva also show up to approximately 50% sequence homology. The characteristic sequence of QGPPP in proline-rich regions of saliva proteins also exists in HRG as Q/HGPPP.

Metal and haem binding to HRG is mediated via the numerous histidine residues within the HRR. Imidazole side

chains of histidine are known to be important for coordination with metal ions.²⁶ Modelling studies predict imidazole binding units (within histidine) that protrude outward from the elongated helical structure of the HRR, allowing access for the binding of metal ions²² (Fig. 2B). At physiological pH, HRG is likely to remain negatively charged, but can acquire a positive charge either through binding Zn^{2+} via non-charged histidine residues, or by direct protonation of histidines under mild acidic conditions,²⁷ a process that causes a conformation change in the molecule that is transmitted through the disulphide bond (Cys399–Cys185) to the N- and C-terminal domains.²² The observation that the interaction between HRG and metals or haem is dependent both on pH and intact histidine residues further supports the idea that the histidine residues constitute the metal binding unit within HRG.²⁸ Even though HRG binds metal ions in the presence of other metal binding serum proteins such as albumin and transferrin, HRG does not appear to play a critical role in serum transport of these ligands under normal physiological conditions. Both in normal human plasma and plasma from a family with elevated Zn^{2+} levels, the majority of Zn^{2+} ions were found to remain bound to albumin and not to HRG.²⁹ It appears more likely that the metal binding function of HRG is to modify the conformation of the molecule and thereby regulate subsequent biological activity.

The cystatin supergene family are more generally known as cysteine proteinase inhibitors (CPI) and are widely distributed in mammalian tissues and body fluids. CPI and HRG share approximately 17% amino acid sequence identity, approximately 32% nucleotide sequence identity, and conserved disulphide bonds in the N-terminal region.³⁰ HRG is often classified as a kininogen-like member of the CPI, although some researchers suggest that HRG should be classified as a new family within the cystatin superfamily.³⁰ HRG also shares N-terminal sequence homology with antithrombin

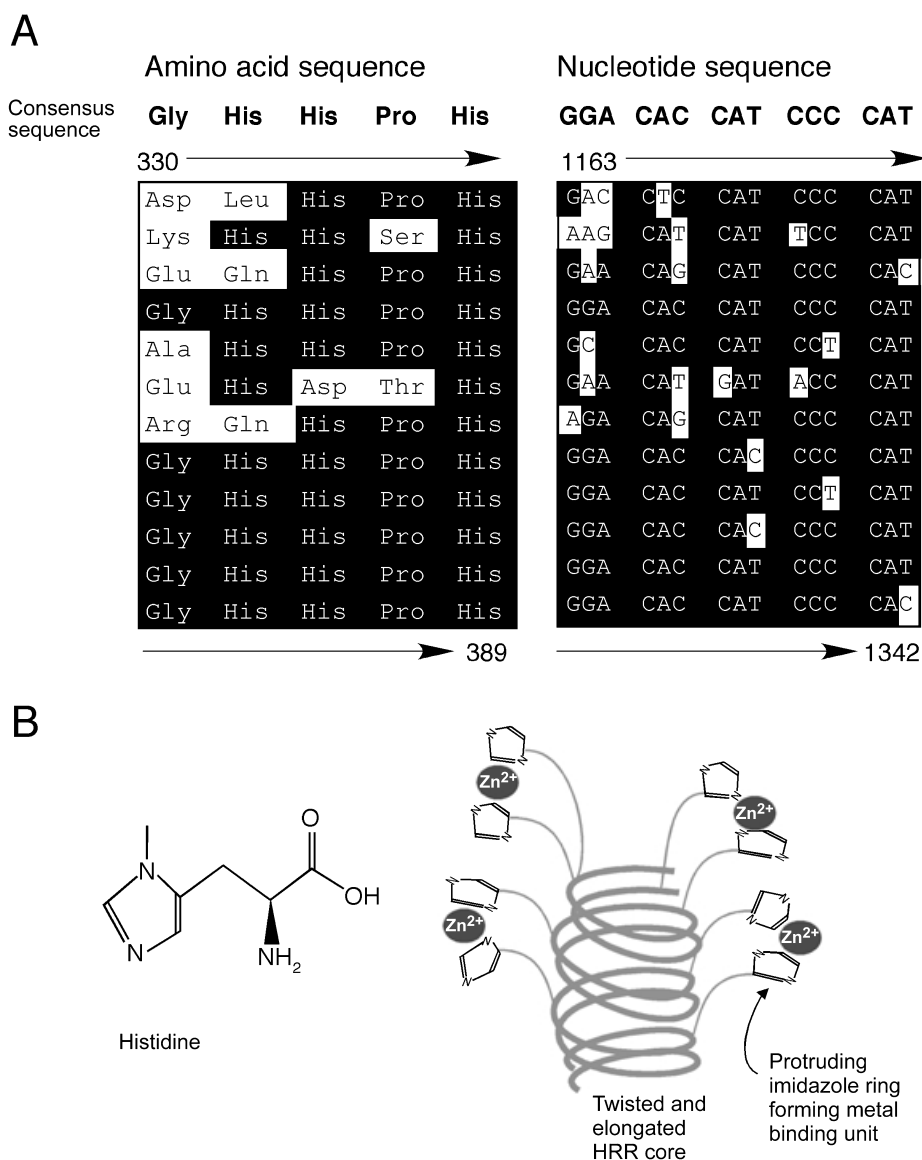


Figure 2 Structure of the histidine-rich region (HRR) of histidine-rich glycoprotein (HRG). (A) Tandem repeats in the HRR of human HRG cDNA and the derived amino acid sequences. Amino acid residues between 330 and 389 are arranged into a block of sequences of five columns and 12 rows, highlighting conservation within the tandem repeat sequences. Amino acids identical with the consensus amino acid sequence ‘Gly-His-His-Pro-His’ are enclosed in a black background. Nucleotides between 1163 and 1342 are arranged into a block of sequences of 15 columns (five codons) and 12 rows. Nucleotides identical with the consensus sequence ‘GGA CAC CAT CCC CAT’ are enclosed in a black background. Adapted from Koide *et al.*³ (B) Structure of a histidine amino acid and projected structure of the HRR of human HRG. With the chemical structure of histidine, note the imidazole ring formed from two nitrogens within the ring structure. Two of these imidazoles form the basic metal binding unit. The helical structure of the HRR of HRG is also depicted, demonstrating that the imidazole side chains protrude from the outer edge of the structure and form the numerous metal binding sites.

III, although no protease inhibitory activity has been reported for HRG.^{19,31}

The secondary structure of HRG has not been entirely determined, however, the N-terminal region appears to be primarily composed of β -sheets, with some α -helix content.²² The hydrophilic HRR generates polyproline (II) helical structures, while the C-terminal domain consists of random coil structures.²² Overall, the predicted secondary structure suggests that the HRG molecule is organized into separate

modular domains rendering numerous multifunctional properties.³ Studies indicate that HRG is readily cleaved by serine proteases such as plasmin from its native form (75 kDa) into fragments ranging in size from 10 to 65 kDa.³² Plasmin cleavage results in the release of the HRR, leaving N- and C-terminal fragments connected by a disulphide bridge between residues 6 and 486,²² which can be separated by further denaturation and reduction. The C-terminal region is easily digested into smaller fragments, whereas the N-terminal

Table 1 Histidine-rich glycoprotein (HRG) ligands

Ligand	$\sim K_d$	Stoichiometry	HRG domain	Reference
Haem	1.5 $\mu\text{mol/L}$	1:1–1:10	HRR	8,43
Zn ²⁺	1–4 $\mu\text{mol/L}$	1:1–1:10	HRR	44
Tropomyosin	2.5 nmol/L	?	HRR	27,90
Plasminogen	0.5–1.1 $\mu\text{mol/L}^\dagger$ 60 nmol/L [‡]	1:1	N-, C-	67,70
Plasmin	0.9 $\mu\text{mol/L}$	1:1	N-, C-	72
Fibrinogen	6.7 nmol/L	?	?	72
Thrombospondin	?	3:1	?	75
IgG	0.6–250 nmol/L	?	N-	80
Fc γ R	?	3:1	N-	83
C1q	7.8 nmol/L	?	N-	81
Heparin and heparan sulphate	7 nmol/L	1:1	N-	48

[†]Soluble HRG; [‡]immobilized HRG.

HRR, histidine-rich region.

domain remains relatively resistant to further proteolysis due to the stabilizing effect of two intradomain disulphide bridges.

HRG gene structure and evolution

The human HRG gene has been mapped to chromosome 3, position 3q28–q29^{33,34} and spans approximately 11 kb. Two other members of the human cystatin gene superfamily, α_2 -Heremans-Schmid (HS) glycoprotein and kininogen also map to this region,³⁵ supporting suggestions that HRG is evolutionarily related to the cystatin gene superfamily. A single gene is predicted to encode HRG in humans and mice,^{3,36} with the human gene consisting of eight exons and seven introns.^{3,33,34,37} Each of the N1 and N2 domains is encoded by three exons, while a single exon encodes the HRR, PRR and C-terminal domain.^{3,34} Several evolutionary implications arise from an analysis of the HRG gene and protein. The many tandem repeats strongly suggest that the HRG gene arose by endoduplication of primordial mini-domains or modules that correspond with previously described type I–V repeats.³⁴ The location of introns within cystatin domains of HRG and members of the cystatin supergene family (cystatin and kininogens) occur in essentially the same positions, suggesting that the HRG gene is evolutionarily related to the cystatin supergene family.^{34,37} Thus, both homologous gene structure and the fact that the HRG and kininogen genes co-localize to a similar region of the human genome supports an evolutionary link between HRG and the cystatin superfamily.³⁷

Two common isoforms of HRG have been found in humans, differing by a molecular weight of approximately 2 kDa due to a polymorphism occurring in exon 5.³⁸ Form-1 HRG (77 kDa) contains a serine residue at amino acid position 186, whereas form-2 HRG (75 kDa) contains a proline residue at position 186. The presence of serine at position 186 introduces an N-glycosylation site at Arg184, making form-1 more highly glycosylated than form-2 and accounting for the 2 kDa difference.³⁸ The frequency of the polymorphisms within the population ($N = 36$) was found to be 0.35 and 0.65 for form-1 HRG and form-2 HRG, respectively.

A novel polypeptide of approximately 85 kDa with a striking similarity to HRG has been isolated from purified human skeletal muscle adenosine monophosphate (AMP) deaminase. AMP deaminase is composed of four subunits,^{39,40}

and contains two different protein species; one is the deaminase catalytic subunit, while the other subunit shows similarity to plasma HRG.⁴¹ A small degree of divergence occurs between AMP-HRG and human HRG in the region containing residues 472–477 and thus the polypeptide is identified as a muscle specific form of HRG, herein referred to as muscle-HRG.⁴¹ It has been suggested that muscle-HRG may be an alternative splice product of the HRG primary transcript, producing a unique N-terminal splice variant.⁴¹ Muscle-specific HRG is clearly localized in type IIB (fast-twitch glycolytic) muscle fibres. It is unknown whether muscle-HRG contributes to the structure of skeletal muscle AMP deaminase or whether it simply binds the enzyme.⁴²

Comparisons between human,³ mouse and rat,³⁶ and rabbit²² HRG amino acid and nucleotide sequences provides insights into conserved regions within HRG that are likely to have an important role in the structure and function of the molecule. Alignment of amino acid sequences from mouse, rat, rabbit and human HRG show a high degree of sequence conservation between species. The overall domain structure of mouse, rat and rabbit HRG is similar to that proposed for human HRG. The PRR are generally the most divergent of the defined regions within HRG. Interestingly, there is very little sequence identity between species in regions that intervene between domains such as between N2 and PRR1 and also between PRR1 and HRR, implying that these regions are unlikely to have an important functional role and may act as a spacer region between the domains.³⁶

HRG ligands

HRG is known to bind a variety of ligands, including haem and divalent cations such as Zn²⁺, tropomyosin, plasminogen, plasmin, fibrinogen, thrombospondin, IgG, Fc γ R, C1q, heparin and heparan sulphate. Table 1 lists the various HRG ligands, indicating the affinity and stoichiometry with which they bind HRG, as well as the domain of HRG with which they interact.

Haem, Zn²⁺ and other divalent metal cations

Early studies revealed that HRG interacts with haem with a stoichiometry of approximately 1:1 to 1:10 and K_d of $\sim 1.5 \mu\text{mol/L}$.⁸ The 12 repeats of the GHHPH motif is approximately equal to

the number of haem binding sites, suggesting that each consensus sequence represents a haem binding unit.⁴³ The haem–HRG interaction is highly dependent on pH, with maximum binding occurring between pH 6.5 and 7.0, precisely when histidine side chains become protonated.^{22,43} Although HRG binds to haem, the physiological relevance of this interaction remains unclear.

HRG interacts with Zn^{2+} with a K_d between 1 and 4 $\mu\text{mol/L}$ and a stoichiometry of approximately 1:1 to 1:10.⁸ Studies indicate the HRR of HRG is most likely the domain responsible for Zn^{2+} binding due to its high level of histidine residues, with two histidine residues forming each metal binding unit. The interaction between HRG and Zn^{2+} is abolished when histidine residues are chemically removed from HRG, supporting the model that histidine residues are essential for Zn^{2+} binding.⁴⁴ Haem and metal ions most likely compete for the same histidine binding unit within the HRR.⁸ It appears that HRG undergoes a conformational change under specific inducible conditions, such as Zn^{2+} binding or low pH, providing a regulatory mechanism for the function of HRG.⁴⁵ HRG interacts *in vitro* with a variety of other divalent metal cations, including copper, mercury, cadmium, nickel and cobalt, with a lower affinity of K_d ($\sim 1 \mu\text{mol/L}$). HRG does not interact with magnesium, manganese or lead.⁴⁴ HRG can also inhibit the formation of spontaneous apatite calcifications formed from calcium and phosphate ions *in vitro*,⁴⁶ presumably by interacting with these divalent cations. Thus, under physiological conditions, it is possible that HRG may accumulate at sites of tissue damage and prevent ectopic calcification by inhibiting apatite precipitation.

Heparin and heparan sulphate

Heparan sulphate proteoglycans are ubiquitously expressed on animal cell surfaces, and in the extracellular matrix (ECM) and vascular basement membrane. Heparin is derived from the heparan sulphate proteoglycans within mast cell granules and is more highly sulphated than heparan sulphate.⁴⁷ HRG interacts with heparin with high affinity, and can neutralize the anticoagulant activity of heparin.^{2,48,49} In purified systems, HRG and heparin interact with a 1:1 stoichiometry to form a high affinity complex with a K_d of $\sim 7 \text{ nmol/L}$.⁵⁰ HRG (in the presence of physiologically attainable concentrations of Zn^{2+}) has the ability to reverse heparin-induced inhibition of smooth muscle cell (SMC) proliferation by binding heparin *in vivo*. Thus, it is possible that HRG may be one of the unidentified factors that reverses the inhibitory effect of heparin on SMC proliferation in arterial microenvironments *in vivo*.^{51,52} In fact, the high affinity of HRG for heparin has resulted in HRG being investigated as an alternative antidote for heparin overdose.⁵³

HRG interacts with heparan sulphate and masks the cleavage site on heparan sulphate chains recognized by the β -D-endoglycosidase heparanase.⁵⁴ Thus, in a physiological environment, soluble HRG may bind to and mask the vascular wall and ECM heparan sulphate, thereby preventing heparan sulphate degradation by heparanase. Heparanase not only cleaves heparan sulphate chains, but also has the important role of regulating the release of growth factors from the ECM. Thus, HRG may block heparanase-mediated liberation of growth factors from heparan sulphate matrices. Such growth

factors potentially play a role in tumour pathogenesis and metastasis. Furthermore, HRG has been shown to bind to cell surface heparan sulphate,^{48,55,56} with maximal binding occurring in the presence of 20 $\mu\text{mol/L}$ Zn^{2+} and/or under conditions of slightly acidic pH.⁴⁸ Although the location of the heparin/heparan sulphate binding site in the HRG molecule has been a matter of debate for many years, a recent study provides direct evidence that the heparin-binding site is located in the N-terminal domain. In this study, a recombinant HRG N-terminal domain was shown to interact with immobilized heparin and cell surface heparan sulphate,⁴⁸ whereas peptides corresponding to the HRR did not bind to cell surfaces. These data are consistent with the earlier observation that the heparin binding site of antithrombin III shares high sequence homology with the N-terminal domain of HRG.^{3,31}

It is well documented that the interaction between HRG and heparin is also dependent on pH, Zn^{2+} and histidine residues.^{48,55–59} Jones *et al.*⁴⁸ suggested a model whereby full-length HRG binds via its N-terminal domain with low affinity to cell surface heparan sulphate in the absence of Zn^{2+} , free Zn^{2+} levels increasing at sites of platelet degranulation to concentrations that promote heparan sulphate binding. Following chelation of Zn^{2+} or protonation of the HRR, a change is induced in HRG that results in the protein binding with higher affinity to cell surface heparan sulphate. Thus, the N-terminal domain of HRG interacts with heparin and heparan sulphate, whereas the HRR modulates this binding by reacting to changes in local pH and Zn^{2+} concentration.

Tropomyosin

Tropomyosin forms part of the actin cytoskeleton⁶⁰ and has been identified on the surface of fibroblast growth factor (FGF)-2-activated human umbilical vein endothelial cells (HUVEC).^{27,61} HRG binds with high affinity ($K_d \sim 2.5 \text{ nmol/L}$) to immobilized tropomyosin in a Zn^{2+} (10 $\mu\text{mol/L}$)- or pH (pH < 6.5)-dependent manner. The HRR domain appears to interact with negatively charged tropomyosin only when the HRR domain acquires a positive charge either through binding to Zn^{2+} via non-charged histidine residues or by direct protonation of histidines induced under mild acidic conditions,²⁷ suggesting that there is a large electrostatic contribution to binding. The levels of free Zn^{2+} in plasma are tightly regulated and are usually too low to promote binding of HRG to endothelial cells.²⁷ Only during coagulation, when activated platelets release free Zn^{2+} from α -granules,⁶² is the local concentration of free Zn^{2+} likely to reach the levels (approximately 10–50 $\mu\text{mol/L}$) needed to support the binding of HRG to tropomyosin on endothelial cells, with this interaction believed to mediate antiangiogenic effects.²⁷

Plasminogen

Plasminogen plays a key role in the fibrinolytic system and is also a HRG ligand. Plasmin activity has been implicated in wound healing, tissue remodelling, angiogenesis, embryogenesis, pathogen and tumour cell invasion, and tumour metastasis.⁶³ Plasminogen contains lysine binding sites that are involved in its interaction with fibrin and HRG.^{18,64–68} The plasminogen binding site within HRG is likely to involve the

N- and C-terminal domains, because immobilized N- and C-terminal domains (but not the HRR domain) can bind plasminogen and stimulate tissue-type plasminogen activator (tPA)-mediated cleavage of plasminogen to plasmin.^{67,69} In particular, lysine residues in the C-terminal domain of HRG are known to be essential for plasminogen binding,^{68,69} although the C-terminal lysine residue does not appear to have an important role in plasminogen binding.⁶⁷

HRG interacts with both plasmin and plasminogen with a K_d of $\sim 1.0 \mu\text{mol/L}$,^{18,70} making HRG one of the most important plasminogen binding proteins in plasma, where it has been estimated that approximately 50% of plasminogen circulates bound to HRG. Plasminogen appears to bind to HRG independently of heparin, indicating that the plasminogen binding sites on HRG are probably distinct from the heparin binding sites.⁶⁸ Numerous studies have confirmed an approximate 1:1 to 1:2 stoichiometry of binding between HRG and plasminogen.^{67,68} Several studies^{67,69,71} have found that immobilized HRG (as opposed to soluble HRG) has a much higher affinity for plasminogen ($K_d \sim 60 \text{ nmol/L}$) resulting in an up to approximately 100-fold enhancement in plasminogen activation by tPA. Interestingly, 70% of the plasmin formed remained surface associated where it was protected from inhibition by α_2 -antiplasmin.⁷¹ These findings are consistent with a physiological model where HRG has a higher affinity for plasminogen when HRG is immobilized on cells via interaction with cell surface heparan sulphate.⁶⁷

Fibrinogen

HRG specifically interacts with fibrinogen with high affinity ($K_d \sim 6.7 \text{ nmol/L}$),⁷² with HRG also being incorporated into fibrin clots with an apparent K_d of $\sim 0.25 \mu\text{mol/L}$. The incorporation of HRG into clots has no effect on the conversion of fibrinogen to fibrin, although interestingly, the presence of HRG in fibrin clots significantly affects the final structure of the fibrin gel by causing the formation of thinner fibrils. The biological significance of HRG interacting with fibrinogen and fibrin may not be limited to the fibrin polymerization process. Fibrin has been shown to form on the surface of thrombin-activated platelets,⁷ and has also been detected on the surface of macrophages,²⁸ thus it is possible that the interaction of HRG with fibrinogen may extend to a role in cell–cell interactions involving platelets and macrophages.⁷²

Thrombospondin

Thrombospondin (TSP) is a 450 kDa filamentous multi-functional α -granule ‘adhesive’ glycoprotein of human platelets that binds to fibrinogen, fibronectin, heparin and HRG.^{73,74} TSP forms a high affinity complex with HRG⁷⁵ and plasminogen ($K_d \sim 35 \text{ nmol/L}$) both via lysine binding sites,^{75,76} and interestingly the HRG–TSP complex is still able to bind plasminogen and heparin. In fact, HRG can form a trimolecular complex with plasminogen, TSP and cell surface heparan.^{75,77} It has been suggested that the complex contains approximately three HRG and approximately three plasminogen molecules per molecule of TSP,⁷⁸ with TSP containing independent binding sites for plasminogen and HRG.⁷⁹ The physiological significance of a surface associated plasminogen–HRG–TSP complex is unclear, although it appears to allow

the efficient activation of plasminogen to plasmin by tPA in microenvironments that do not contain fibrin, and thus potentially facilitates plasmin-mediated physiological and pathological events such as ovulation, implantation, inflammation and tumour cell invasion.⁷⁶

IgG, Fc γ R and complement

Immunoglobulin (Ig) molecules consist of two identical heavy (H)- and light (L)-chains. In many species, two types of L-chain exist, termed κ and λ , which exhibit little amino acid homology and, unlike H-chains, their function remains unknown. Gorgani *et al.*⁸⁰ showed that HRG can interact with soluble IgG, although the affinity of this interaction is enhanced up to fivefold by Zn^{2+} ($20 \mu\text{mol/L}$), is dependent upon the IgG subclass, and is influenced by whether the IgG molecule contains κ or λ L-chains. Thus, IgG1 and IgG2 κ L-chain containing forms have a 50–200-fold greater affinity for HRG than the IgG1 and IgG2 λ L-chain forms, providing the first evidence for a functional difference between these two L-chains. The HRG–IgG interaction is thought to occur predominantly via the $F_{(ab)}$ region of IgG, and the N-terminal domain of HRG.⁸¹ It has also been reported that HRG interacts with Fc receptors, suggested to be Fc γ RI and Fc γ RII, via its N-terminal domain and possibly C-terminal domains in a Zn^{2+} -independent manner.^{82,83}

HRG was first discovered by Heimburger *et al.*² when it copurified with the first component of complement (C1q). The N-terminal domain of HRG interacts with C1q via two distinct high affinity binding sites ($K_d \sim 7.8 \text{ nmol/L}$ and $\sim 37 \text{ nmol/L}$).⁸¹ Interestingly, Zn^{2+} ($20 \mu\text{mol/L}$) appears to inhibit the interaction between HRG and C1q. The inhibitory effect of Zn^{2+} may be due to an induced conformational change in the HRG molecule resulting in a loss of C1q binding sites. Immobilized HRG also binds many of the membrane attack complex (MAC) components,⁸⁴ suggesting that HRG may play a role in regulating the formation of the MAC and modulating complement functional efficiency.

HRG cell surface receptors

HRG binds many cell types, including T lymphocytes,^{85,86} murine erythrocytes,⁸⁷ EBV-transformed human B cells⁸⁶ and platelets^{88,89} as well as endothelial,²⁷ melanoma, ovarian,⁶⁷ fibroblast,⁵⁵ lymphoma⁵⁶ and myeloid cell lines.⁸⁰ Despite the ability of HRG to bind to a diverse range of cell types, HRG cell surface receptors in many cases are not clearly defined. Nevertheless, four broad classes of cell surface receptors have been identified: (i) heparan sulphate, (ii) tropomyosin, (iii) an unidentified 50–55 kDa protein on T cells and (iv) FcR.

Heparan sulphate glycosaminoglycans are ubiquitously expressed on most cells surfaces. Most HRG cell surface binding appears to be mediated via cell surface heparan sulphate. Indeed, early studies by Parish *et al.*⁸⁷ concluded that human HRG interacts with anionic carbohydrates, probably heparan sulphates, on murine erythrocytes. Jones *et al.*⁴⁸ provided clear evidence that HRG interacts with cell surfaces via heparan sulphate, with treatment of cells with mammalian heparanase destroying HRG binding sites. Similarly, HRG was shown to bind to CHO-K1 cells, but not to a mutant form of CHO-K1 cells that lacks cell surface heparan sulphate,

providing further support for heparan sulphate being one of the primary HRG cell surface ligands. In addition, Guan *et al.*²⁷ showed that HRG binds to heparan sulphate proteoglycans expressed on the surface of HUVEC.

Recent studies have identified tropomyosin as a novel cell surface receptor for HRG.^{27,90} Fibroblast growth factor (FGF)-2 stimulation of HUVEC causes translocation of tropomyosin to the cell surface. Guan *et al.*²⁷ observed that HRG binds with higher affinity to FGF-2 stimulated HUVEC ($K_d \sim 6$ nmol/L) than to non-stimulated cells ($K_d \sim 55$ nmol/L), with subsequent work revealing that tropomyosin is the high affinity binding site for HRG on FGF-2 stimulated HUVEC. Interestingly, the binding of HRG to immobilized tropomyosin and FGF-2 stimulated HUVEC requires Zn^{2+} at pH 7.5.²⁷ However, binding can occur in the absence of Zn^{2+} at pH 6.0. Thus, at physiological pH, binding is dependent on the presence of Zn^{2+} whereas binding can also occur in the absence of Zn^{2+} at slightly acidic pH (pH < 6.5).

Studies with human peripheral blood T lymphocytes suggests that heparan sulphate-independent binding of HRG to cell surfaces can occur.⁸⁶ Kinetic analyses suggested both high affinity ($K_d \sim 19$ nmol/L) and low affinity ($K_d \sim 500$ nmol/L) binding sites on human T cells for HRG. Furthermore, unlike HRG binding to most other cell types, the presence of Zn^{2+} did not significantly enhance HRG binding to T lymphocytes.⁸⁶ An unidentified 50–55 kDa protein isolated from T lymphocyte lysates was identified as the specific HRG T-cell surface receptor. More recent studies by Olsen *et al.*⁵⁶ further investigated HRG binding to T cells using human and mouse T cell lines, and found that Zn^{2+} profoundly potentiated HRG binding to T lymphocytes, and that heparin inhibited binding, contrary to the results of Saigo *et al.*⁸⁶ Thus, perhaps heparan sulphate may be a more important ligand on activated and/or transformed T lymphocytes⁵⁶ than on resting peripheral blood T lymphocytes.⁸⁶

Finally, Fc receptors have been shown to be a ligand for HRG on a number of different cell types. Chang *et al.*⁹¹ demonstrated that HRG binds to inflammatory peritoneal macrophages via the low affinity Fc receptor, Fc γ RII, whereas studies by Gorgani *et al.*⁸² suggested that HRG can interact with Fc γ RI, a high affinity Fc receptor on myeloid cells.

Functional effects of HRG

pH sensor and Zn^{2+} detector

Numerous studies indicate that the interaction between HRG and Zn^{2+} is highly dependent on pH. Morgan⁴⁴ hypothesized that the unique, conserved HRR domain of HRG acts as a pH sensor, providing a mechanism for regulating HRG activity. In their model, histidine side chains of HRG become protonated when local pH falls during conditions such as ischemia or hypoxia. The HRR becomes positively charged, allowing HRG to subsequently strongly bind negatively charged GAG on cell surfaces. This pH sensor phenomenon may become physiologically relevant as it allows HRG to function as an adaptor molecule, co-immobilizing soluble proteins, such as plasminogen, onto cell surface GAG only under specific conditions. In support of this concept, *in vitro* studies showed that immobilized, but not soluble, HRG stimulates plasminogen activation.⁶⁹ Thus, in a physiological setting, cell surface GAG can bind HRG *in vivo* when the local pH is lowered, a

process that may result in plasmin generation within a confined and regulated local environment. Subsequent studies⁶⁷ have confirmed that HRG can target plasminogen/plasmin to the surface of cells under conditions of elevated free Zn^{2+} (~ 20 μ mol/L) or low pH (pH < 6.5). Because plasmin has an important role in aiding ECM degradation by invading cells, HRG may potentiate leucocyte entry into inflammatory sites, angiogenesis and tumour metastasis via this mechanism.

The pH dependence of the HRG–GAG interaction is likely to be physiologically relevant as the local pH can drop by as much as one pH unit under conditions of hypoxia or ischemia,^{92,93} or 0.5 of a pH unit during an inflammatory response due to lactic acidosis.⁹⁴ Thus, HRG may function as a pH sensor that binds cell surface GAG only when local pH decreases or when local Zn^{2+} increases, such as during tissue injury or tumour metastasis.^{45,67} Due to the HRR of HRG acquiring a positive charge upon protonation, HRG can be described as a facultative polycation.⁶⁹ It seems likely that many of the biological actions of HRG are related to its polycationic nature, implying that the functional effects of HRG are regulated by pH and expressed only under physiological conditions that produce local acidosis.

Modulator of cell proliferation and cell adhesion

Studies suggest that HRG modulates cell proliferation by interacting with cell surface GAG. HRG has been reported to induce proliferation of cultured arterial smooth muscle cells by inhibiting the antiproliferative effects of heparin.^{51,52} Other studies however, indicate that HRG inhibits cell proliferation. Shatsky *et al.*⁸⁵ showed that HRG was capable of suppressing T-cell proliferation. Brown and Parish⁵⁵ also showed that HRG inhibits proliferating fibroblasts by interacting with cell surface heparan sulphate and subsequently inhibiting the binding of acidic and basic FGF to these cells. These findings support a role for HRG in modulating the release of FGF from the ECM and basement membranes and also in inhibiting the interaction of FGF with cell surface heparan sulphates. Thus HRG has been reported to both inhibit and enhance cell proliferation by interacting with heparan sulphate on cell surfaces.

Long-term treatment of peritoneal macrophages with physiological concentrations of HRG results in the loss of cell adherence, probably mediated via reduced synthesis of cell surface adhesive proteins.⁹¹ Olsen *et al.*⁵⁶ also found that HRG inhibits the adhesion properties of murine T-cell lines to plastic culture dishes, but that HRG promoted homotypic adhesion between the cells. In contrast, Lamb-Wharton and Morgan⁹⁵ reported that HRG promotes T-cell adhesion, whilst also inducing a change in morphology. HRG induced MOLT-3 cells either to produce extended cellular processes or to become elongated at the poles. The changes in adherence status and morphology observed in MOLT-3 cells are similar to those of leucocytes preparing to migrate through the micro-vascular system to sites of inflammation, and suggest that HRG may be involved in regulating early adhesion events of T lymphocytes during immune responses and at sites of inflammation.⁹⁵ In addition, HRG has the ability to inhibit autorosetting (cell adhesion between autologous murine lymphocytes and erythrocytes), and thus HRG is also known as 'autorosette inhibition factor'.^{6,96–98}

Table 2 Multiple effects of histidine-rich glycoprotein (HRG) on angiogenesis

Enhancing effects
Over-rides the antiangiogenic effects of thrombospondin-1 ^{113†}
May tether plasminogen/plasmin to the surface of endothelial cells and aid cell migration and invasion ^{67‡}
Inhibitory effects
Blocks FGF-1 and FGF-2 action by preventing FGF binding to heparan sulphate in the ECM and on endothelial cell surfaces ^{55‡}
Prevents heparanase-mediated release of angiogenic growth factors from the ECM by masking heparanase cleavage sites in ECM heparan sulphate ^{54‡}
Delivers an antiangiogenic signal to endothelial cells by binding to cell surface tropomyosin via its HRR ^{27,90†,‡}

[†]Probably an angiogenesis-specific effect; [‡]processes that are likely to be enhanced by elevated extracellular free Zn²⁺ levels (usually provided by degranulating platelets) and by acidosis (pH < 6.5).

ECM, extracellular matrix; FGF, fibroblast growth factor; HRR, histidine-rich region.

Regulator of coagulation and fibrinolysis

HRG has been found to modulate various components of the coagulation cascade. HRG can bind to heparin (a potent anti-coagulant), resulting in a failure of mast cell-released heparin to inhibit the procoagulant activity of monocytes at sites of inflammation and thrombosis.^{99,100} Clinical studies also support an interaction between HRG and heparin *in vivo*, HRG blocking the normal interaction between heparin and anti-thrombin III.^{78,101,102} Other studies have investigated whether HRG can act as an antifibrinolytic agent due to its high affinity interaction ($K_d \sim 0.5\text{--}1.1 \mu\text{mol/L}$) with plasminogen. HRG reportedly binds to lysine binding sites on plasminogen, thus preventing plasminogen from interacting with fibrinogen and α_2 -antiplasmin. Although there is no direct evidence to support this view, it has been proposed by numerous groups that HRG inhibits the interaction of plasminogen with fibrin clots, thus indirectly inhibiting plasmin mediated fibrinolysis.¹⁸ This proposed antifibrinolytic effect of HRG would be expected to potentiate clot formation and persistence in the vasculature and in tissues. In contrast to the antifibrinolytic effect of HRG, other studies^{69,71} have demonstrated that surface immobilized HRG can act as a pro-fibrinolytic agent by enhancing fibrinolysis. Immobilized HRG binds plasminogen with high affinity resulting in an approximately 30-fold increase in the conversion of plasminogen to plasmin by tPA, with theoretically a concomitant increase in fibrinolytic potential in the microenvironment of fibrin clots.⁶⁹ Collectively, these data suggest that HRG can regulate coagulation and fibrinolysis, but whether this represents a major function of HRG *in vivo* remains to be established.

Modulator of immune complex clearance and Fc γ R-mediated phagocytosis

Certain autoimmune diseases such as systemic lupus erythematosus (SLE) are associated with the production of insoluble immune complexes (IIC) and consequent tissue injury.^{103,104} Gorgani *et al.*^{81,105} found that physiological concentrations of the N-terminal domain of HRG can inhibit the first stage of formation of IIC,¹⁰⁶ suggesting that HRG may have a role in regulating IIC formation *in vivo*. HRG probably inhibits IIC formation by preventing Fc–Fc interactions and by masking epitopes on IgG recognized by rheumatoid factors.^{81,105} Although HRG was also found to promote solubilization of already formed IIC, HRG was more effective at inhibiting the

formation of IIC. Hence, evidence suggests that HRG may have an important role in the formation, solubilization and subsequent clearance of IIC *in vivo*.

HRG has also been shown to biphasically modulate the expression of Fc γ R and its associated phagocytic function by murine peritoneal macrophages. After 1–2 h of HRG treatment, Fc γ R expression is rapidly upregulated on inflammatory peritoneal macrophages, resulting in enhanced phagocytosis of opsonized sheep erythrocytes, but after 16–48 h of treatment, Fc γ R expression is downregulated, resulting in significantly reduced phagocytosis of opsonized sheep erythrocytes.^{83,91} Both the N-terminal and C-terminal domains of HRG appear to be important for regulating macrophage phagocytic function.⁸³

Recent studies by Gorgani *et al.*¹⁰⁷ demonstrated that HRG can increase the rate of phagocytosis of apoptotic cells by human monocyte derived macrophages (HMDM). Based on these observations, the authors postulated that HRG forms a bridge between monocytic and apoptotic cells via Fc γ RI on HMDM and naked DNA on apoptotic cells. Thus, HRG may assist in the removal of apoptotic cells under normal physiological conditions, whereas other proteins such as C-reactive protein take over during more pathological or chronic inflammatory disorders.¹⁰⁷ Gorgani *et al.*¹⁰⁷ suggest that HRG may protect individuals from the development of systemic autoimmunity by promoting the efficient removal of apoptotic cells. This theory is supported by HRG levels being low in the sera of human SLE patients.¹⁰⁸ However, future work with HRG knockout mice and mice predisposed to SLE should provide more data to evaluate this hypothesis.

Regulator of angiogenesis

It has been reported that HRG exhibits both enhancing and inhibitory effects on angiogenesis, which are summarized in Table 2; some of these effects are highly specific to angiogenesis. During angiogenesis, new capillary blood vessels are formed from pre-existing vessels in response to angiogenic stimuli. TSP is known to be a potent inhibitor of angiogenesis by binding to a CLESH-1 (CD36, LIMPII, *emp*, SR-BI homology sequence 1) motif in CD36.^{109,110} The TSP sequence 'CSVTCG' is thought to be responsible for delivering the antiangiogenic signal to CD36, this peptide having been shown to possess antiangiogenic activity as a free peptide.^{111,112} There are two regions of HRG with significant homology to the CLESH-1 motif, between amino acid

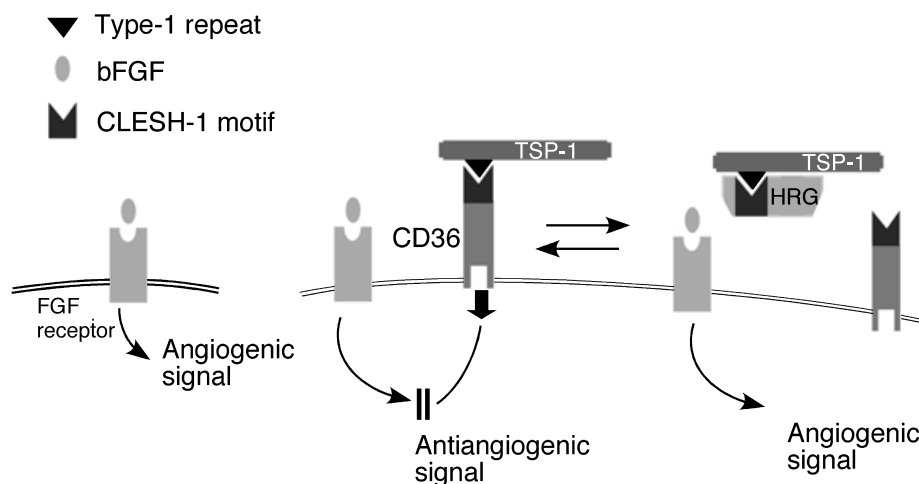


Figure 3 Model of inhibition of the antiangiogenic effect of thrombospondin (TSP)-1 by histidine-rich glycoprotein (HRG). Angiogenesis induced by basic fibroblast growth factor (bFGF) is inhibited by TSP-1 through the interaction of the TSP type-1 repeat with the CLESH-1 domain of the signalling receptor CD36 (centre). HRG also contains a CLESH-1 motif and upon binding TSP-1, inhibits the interaction of TSP-1 with CD36, thereby inhibiting the antiangiogenic effect of TSP (right). Adapted from Simantov *et al.*¹¹³

residues 155–213 in the N2 domain, and between residues 428–503 in the C-terminal domain.¹¹³ Simantov *et al.*¹¹³ proposed a model in which HRG inhibits the antiangiogenic effects of TSP by blocking the interaction between TSP and CD36, thereby preventing delivery of the antiangiogenic signal (Fig. 3). Thus, HRG may act as a natural modulator of angiogenesis by antagonizing the effect of endogenous TSP.¹¹³ Interestingly, HRG may also provide a mechanism by which tumours escape or become resistant to the antiangiogenic effects of TSP. Immunohistochemical studies of breast cancer specimens show HRG co-localizes with TSP in the tumour matrix and that the interaction masks the antiangiogenic epitope of TSP. Therefore, in areas where TSP is an important inhibitor of angiogenesis, HRG may serve as a modulator of TSP activity by promoting angiogenesis.¹¹³ Furthermore, it has also been reported that HRG may tether plasminogen/plasmin to the surface of cells and potentially aid endothelial cell migration and invasion during angiogenesis, although this phenomenon may also aid the invasion of many different cell types.⁶⁷

Conversely, other studies indicate that HRG exhibits antiangiogenic activity by various mechanisms.^{27,54,55,90,114,115} First, HRG binds to heparan sulphate on the surface of endothelial cells and prevents the heparan sulphate serving as a coreceptor for various FGF.⁵⁵ Clearly this inhibitory effect is not restricted to angiogenesis, but could also block FGF-dependent proliferation of many cell types. Second, HRG prevents heparanase-mediated release of angiogenic growth factors, such as FGF, from the ECM by masking heparanase cleavage sites in ECM heparan sulphate.⁵⁴ Again, this inhibitory effect is probably not restricted to angiogenesis. Third, the HRR of HRG has been shown to deliver antiangiogenic signals to endothelial cells.^{27,90,114,115} Initially HRG was shown to inhibit tube formation and the proliferation of endothelial cells,¹¹⁴ as well as blocking angiogenesis in *in vivo* models.²⁷ The antiangiogenic effect of HRG appeared to be mediated specifically through the GHHPH consensus region within the

HRR.⁹⁰ In addition, Olsson *et al.*¹¹⁵ showed that treatment of fibrosarcoma-bearing mice with HRG resulted in decreased tumour angiogenesis and a decrease in tumour volume. Similarly, peptides derived from the HRR have also been shown to inhibit angiogenesis and tumour growth *in vivo*.⁹⁰ Additional studies also identified tropomyosin to be the key HRR binding molecule on human endothelial cells, with binding being optimal under conditions of elevated Zn^{2+} and/or acidic pH (Fig. 4).²⁷ Because tropomyosin lacks a transmembrane domain, it is likely that other proteins in the endothelial cell membrane may associate with tropomyosin and play a role in the antiangiogenic signalling. It has been suggested that HRG mediates its antiangiogenic activity by inducing apoptosis in activated endothelial cells and reducing cell proliferation.¹¹⁴ Olsson *et al.*¹¹⁵ also suggested that HRG acts by decreasing the attachment of endothelial cells to vitronectin and reducing endothelial cell migration. The potent antiangiogenic effects of specific peptide sequences derived from the HRR of HRG are currently being pursued for possible development as antiangiogenic agents.⁹⁰

Conclusion

HRG is a multidomain molecule that interacts with many ligands. The multidomain structure of HRG suggests that the molecule may act as an adaptor protein that can bring together different ligands under specific conditions. Furthermore, the HRR of HRG is highly responsive to Zn^{2+} and pH, which causes a conformation change within the HRR that can be transmitted to other domains of the molecule to regulate their function. The HRR is also the domain of the molecule that can interact with tropomyosin on tumour cells, resulting in the production of an antiangiogenic signal. Thus, although HRG has been shown to interact with a variety of ligands, and to regulate numerous biological activities, much more remains to be discovered about this intriguing molecule.

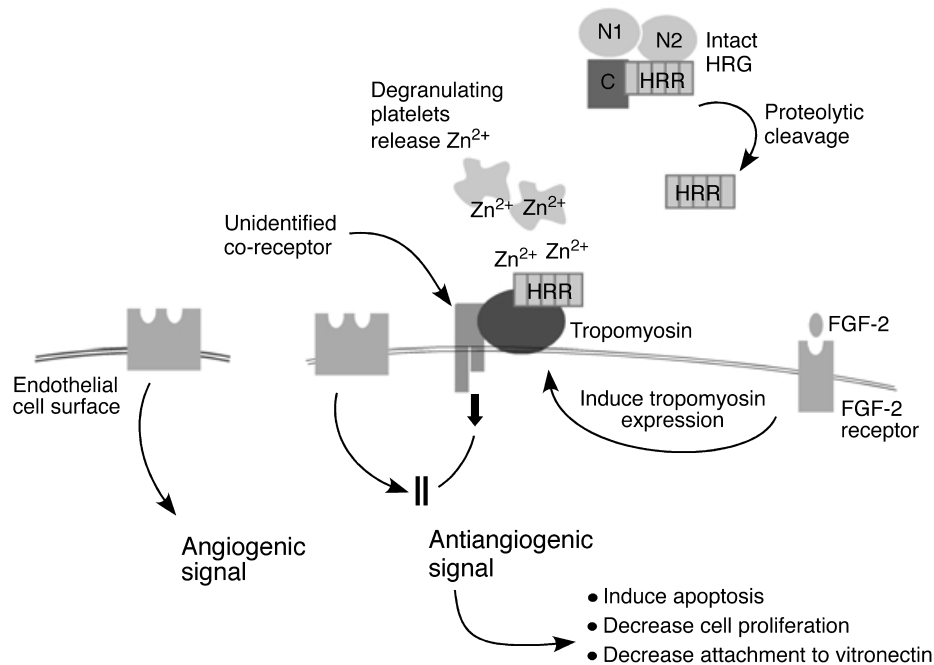


Figure 4 Model of the antiangiogenic effect mediated via the histidine-rich region (HRR) of histidine-rich glycoprotein (HRG). Fibroblast growth factor (FGF)-2 binds to its cell surface receptor and stimulates higher levels of expression of tropomyosin on the surface of endothelial cells. The HRR domain of HRG then binds with high affinity to cell surface tropomyosin and delivers an antiangiogenic signal to the endothelial cells. Tropomyosin lacks a transmembrane domain, suggesting that it may require a coreceptor to mediate the antiangiogenic signal. The binding of the HRR to tropomyosin is dependent on pH and Zn²⁺. At physiological pH, binding of the HRR to tropomyosin is dependent on the presence of Zn²⁺, with local elevated free Zn²⁺ concentrations (10–50 µmol/L) being provided by degranulating platelets, although binding can also occur in the absence of Zn²⁺ at slightly acidic pH (pH = 6.5). The antiangiogenic signal causes increased apoptosis, decreased cell proliferation and cell migration. This schematic diagram was based on results discussed in Donate *et al.*,⁹⁰ Guan *et al.*²⁷ and Olsson *et al.*¹¹⁵

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